LEC32 - Exoplanet Statistics

Friday, November 29, 2024

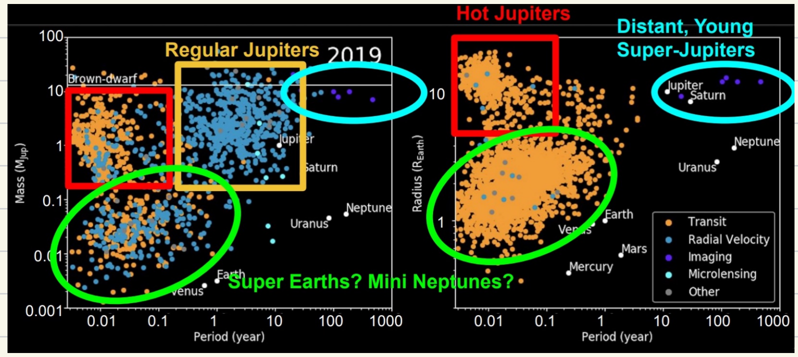
Learning Goals

- ☐ Describe the statistics of planetary occurrence
- ☐ Compare the occurrence of giant planets at different orbital separations
- ☐ Describe the properties of small planets → mini-Neptunes and super-Earths
- ☐ Discuss the occurrence rates of "habitable" planets

Survey sensitivity: What fraction of planets could we detect with our data?

Demographics of Exoplanets

- Diagrams only show raw numbers
- Some planets are easier to detect than others



Calculating Occurrence Rates

- $\text{Occurence} = \dfrac{\text{Planet Found} + \text{Planets Missed}}{\text{Number Stars}}$

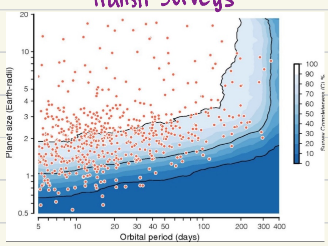
Occurrence vs. Planet Types

- Hot Jupiters → < 1%
- Super-Earths + Mini-Neptunes → ~100%

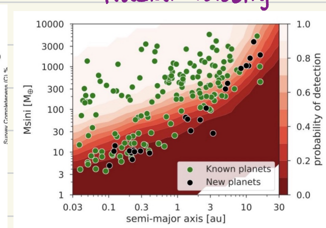
With < 1 year orbits

- Giant planet occurrence rates tell us about formation
  - → Occurrence rate peaks ~3AU
    - Near water frost line

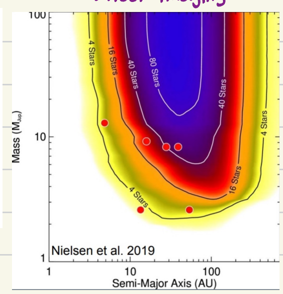
Transit Surveys



Radial Velocity

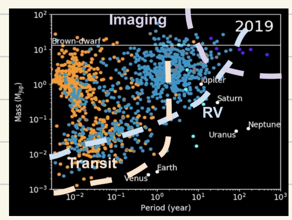


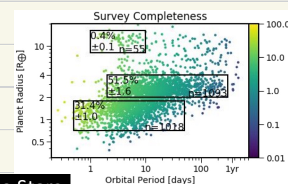
Direct Imaging

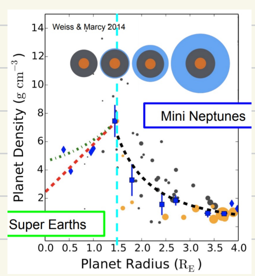


Small Planets: Super-Earths and mini-Neptunes

- Density peaks around $1.5R_{earth}$
- $R < 1.5R_{earth}$
  - → Rocky like Earth, with minimal atmosphere
- $R > 1.5R_{earth}$
  - → Probably rocky cores with voluminous hydrogen/helium atmospheres



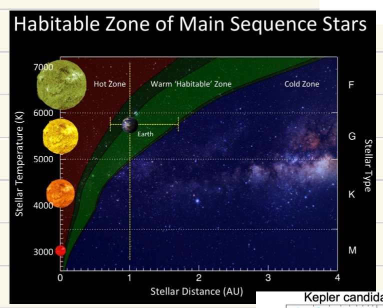


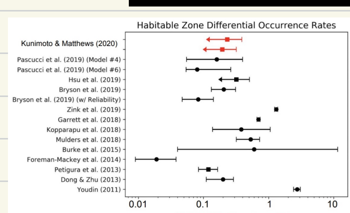


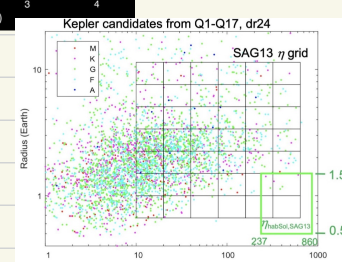
Habitable Planets

- Possible requirements for surface habitability
  1. Moderate temperature
     - → To support liquid water on surface
     - → Sets distance to star (recieved flux)
  2. Volcanism
     - → To make and maintain atmosphere
  3. Plate tectonics
     - → To regulate the climate via CO_2 cycle
  4. Magnetic Field
     - → To protect the atmosphere from stellar radiation

Sets planet size: large enough to retain heat, small enough to be rocky







How common are planets in the habitable zone?

- Highly dependent on definition of habitability
- Planet size range
  - → Large enough to maintain plate tectonics and atmosphere ($>0.5R_E$)
  - → Small enough to be rocky ($<1.5R_E$)
- Stellar Irradiation
  - → Hard to determine planets temperature without knowing about its greenhouse/albedo
  - → Rough estimate is 0.5 - 2x flux recieved by Earth
  - → More detailed models of Venus and Mars to set boundaries assuming atmospheric composition